

Rocketry Research paper

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1 Introduction

I am Maxwel Kim, and I will be writing this research paper for the purpose of revision, data analysis and records of my work + career. This will explain how each physics terms, formulas and my self-studied advanced mathematics material will be used for my rocketry club. Unlike most students, I use "rocketry" as an application for my knowledge of mathematics rather than focusing on building it to achieve certain height.

Predicting rocketry apogee is a fundamental problem since accurate apogee estimates are important for vehicle design, flight safety and recovery. The app OpenRocket simulates trajectories with high accuracy, yet these simulations lack mathematical model, insight into how launch parameters influence the final altitude.

Existing models such as those in OpenRocket are primarily by numerical simulations and empirical predictions. While these work out accurately on estimation, they do not reveal the sensitivity of the prediction that changes directly in launch conditions. They also do not provide mathematical model/framework for analyzing optimization, geometric structure.

this paper studies an empirical apogee model that depends on rocket weight, wind conditions(directions, magnitude), and ambient temperature. rather than solving solely on prediction, the model has been investigated using various techniques such as multivarious calculus, differential equations, optimization, differential geometry, and topology.

The primary contributions of this work are formulation of a mathematical framework for the empirical apogee model, local sensitivity analysis using Taylor expansions and gradients, curvature analysis using the Hessian, a geometric interpretation of launch conditions as a

smooth manifold, topological analysis of constant-apogee level sets, validation against numerical simulation.

The remainder of this paper is organized as follows. Section 2 presents the physical motivation and empirical model. Section 3 estimates model parameters using regression. Section 4 develops the Taylor expansion and sensitivity analysis. Sections 5–7 introduce differential geometry, topology, and differential equations. Section 8 presents numerical validation, and Section 9 concludes the paper.

2 Physical Background

2.1 Newton’s second law

The motion of rocket is governed by Newton’s Second Law,

$$m\ddot{x} = T - D - mg$$

where m is the rocket mass, T is the thrust produced by the motor, and D is the aerodynamic drag force, g being the gravitational acceleration. The thrust term explains the force generated by the expulsion of propellant, accelerating the rocket upward. The drag force represents aerodynamic resistance that is caused by interaction between the rocket and the surrounding air. the gravitational term opposes upward motion and depends on the rocket’s mass.

rearranging the equation by diving both sides by m , we get:

$$\ddot{x} = \frac{T - D}{m} - g$$

this is significant because this shows how acceleration depends on the balance between propulsion, aerodynamic thrust, and rocket mass. Since all those term vary throughout the flight, obtaining an accurate apogee result becomes difficult to achieve. This therefore motivates the use of approximation by mathematical model.

2.2 aerodynamic drag

the aerodynamic drag force opposing rocket motion is modeled as:

$$D = \frac{1}{2}\rho C_D A v^2$$

where ρ is air density, C_D is drag coefficient, A is reference cross-sectional area, and v the velocity that is relative to surrounding air.

Drag increases quadratically with velocity, making it one of the most sensitive variables during the flight. the magnitude therefore is not only for geometric

expression but also atmospheric conditions. In particular, changes in air density caused by temperature variations alter the aerodynamic resistance experienced by the rocket. Since wind changes the relative velocity between the rocket and the air, it also influences the effective drag force. Therefore, temperature and wind conditions indirectly affect the final apogee height.

2.3 Atmospheric Density and Temperature

The aerodynamic behavior of a rocket is strongly influenced by the surrounding atmosphere. The relationship between air density, pressure, and temperature is described by the ideal gas law,

$$\rho = \frac{P}{RT},$$

where ρ is the air density, P is atmospheric pressure, R is the specific gas constant, and T is the absolute temperature.

As temperature increases, air density decreases, reducing the aerodynamic drag force acting on the rocket. Conversely, lower temperatures increase air density and produce greater drag. Since drag directly affects the acceleration and velocity of the rocket, temperature variations influence the final apogee.

For small changes around a reference temperature T_0 , the effect of temperature on air density can be approximated using a Taylor expansion,

$$\rho(T) \approx \rho(T_0) + \left. \frac{d\rho}{dT} \right|_{T_0} (T - T_0),$$

where

$$\Delta T = T - T_0.$$

Therefore, the first-order temperature contribution to apogee can be modeled as approximately linear,

$$f_T(\Delta T) = B\Delta T.$$

Where B here is a constant.

2.4 Wind Effects

As we talked earlier, looking at Aerodynamic drag force equation, the drag itself is very sensitive to what relative velocity is, and we define relative velocity as

$$\vec{v}_{rel} = \vec{v}_r - \vec{v}_w$$

where each notation stands for relative velocity being subtraction from rocket's velocity to wind's.

The magnitude of the relative velocity is

$$v_{rel} = |\vec{v}_r - \vec{v}_w|,$$

which directly affects aerodynamic drag through

$$D = \frac{1}{2} \rho C_D A v_{rel}^2.$$

For a wind vector making an angle θ with the rocket trajectory, the component of wind along the flight direction is given by

$$V_{\parallel} = V_w \cos(\theta),$$

where $V_w \cos(\theta)$ is the wind speed.

This dependence from effective velocity and drag force motivates including a wind related term in the apogee model.

2.5 Mass Effects

The mass of the rocket is a fundamental parameter affecting its acceleration and resulting apogee. From Newton's second law,

$$a = \frac{F}{m},$$

where a is acceleration, F is the net force acting on the rocket, and m is the rocket mass.

For a fixed propulsion system, increasing the rocket mass reduces the acceleration produced by the motor. Since the velocity gained during powered flight determines the initial conditions for the coast phase, changes in mass directly influence the maximum altitude.

The net acceleration can be expressed as

$$\ddot{x} = \frac{T - D}{m} - g.$$

This relationship demonstrates that mass affects both the gravitational force and the acceleration response to thrust and drag. Therefore, rocket mass is expected to have a significant influence on apogee.

For nonlinear effects over a range of masses, a logarithmic transformation can be used to represent diminishing sensitivity,

$$f_W(W) = C \ln \left(\frac{W}{W_0} \right),$$

where W_0 is a reference mass and C is a model coefficient.

3 Mathematical Model

3.1 Model Formulation

The previous section demonstrates that rocket apogee depends on several interacting physical variables, including rocket mass, wind conditions, and atmospheric temperature. Due to the complexity of the governing equations and the changing nature of aerodynamic forces during flight, an exact analytical solution for apogee is difficult to obtain.

Therefore, the apogee is modeled as a function of the dominant environmental and physical parameters,

$$H = F(W, V, \Delta T),$$

where H represents the predicted apogee, W represents rocket mass, V represents wind conditions(vectors), and ΔT represents the temperature change from reference condition.

For small variations around a nominal launch condition, the apogee function can be approximated by decomposing the contribution of each variable,

$$H \approx A + f_W(W) + f_V(V) + f_T(\Delta T),$$

where A is the baseline apogee and the remaining terms represent the individual contributions of each variable.

The mass contribution is modeled using a logarithmic relationship,

$$f_W(W) = C \ln \left(\frac{W}{W_0} \right),$$

to represent diminishing sensitivity to changes in mass.

The temperature contribution is approximated linearly,

$$f_T(\Delta T) = B\Delta T,$$

based on the first-order Taylor approximation of atmospheric effects.

The wind contribution is modeled through the velocity component along the flight direction,

$$f_V(V) = D(V_w \cos \theta).$$

Therefore, the complete empirical model becomes

$$H = A + B\Delta T + C \ln \left(\frac{W}{W_0} \right) + D(V_w \cos \theta)$$

The coefficients A, B, C, D represent the sensitivity of apogee to each variable.

A = baseline apogee

B = temperature sensitivity

C = mass sensitivity

D = wind sensitivity

These parameters can be determined using numerical simulation or experimental flight data.

4 Parameter Estimation

4.1 Data Generation

To determine the coefficients of the apogee model, numerical flight simulations are performed under varying launch conditions. Each simulation produces an apogee value corresponding to a specific combination of rocket mass, wind conditions, and temperature variation.

The dataset is represented as

$$\mathcal{D} = \{(W_i, V_i, \Delta T_i, H_i)\}_{i=1}^n,$$

where n represents the number of simulated flights.

The input variables are varied within realistic launch conditions, while the resulting apogee values are used as the target output for model fitting.

Trial	Apogee (ft)	Weight (g)	Temperature (°F)	Wind (mph)
1	753	575	62.6	4.35
2	773	576	60.8	4.35
3	764	586	60.8	7
4	716	603	60.8	6
5	837	560	40	2
6	813	583	43	2
7	731	624	45	2

Table 1: Successful flight data used for regression analysis.

4.2 Regression Model

The empirical apogee equation can be expressed as a linear regression model by defining transformed variables,

$$x_1 = \Delta T,$$

$$x_2 = \ln \left(\frac{W}{W_0} \right),$$

$$x_3 = V_w \cos \theta.$$

Therefore,

$$H = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

4.3 Least Squares Solution

The coefficients are estimated by minimizing the squared error,

$$\min_{\beta} \|X\beta - H\|^2.$$

The least squares solution is

$$\hat{\beta} = (X^T X)^{-1} X^T H$$

which provides estimates for A, B, C, D .

4.4 Model Coefficients

Applying the least squares estimator to the experimental dataset gives the coefficient vector

$$\hat{\beta} = \begin{bmatrix} 794.1 \\ -3.12 \\ -920.8 \\ 3.18 \end{bmatrix}.$$

Therefore, the fitted apogee model is

$$H = 794.1 - 3.12\Delta T - 920.8 \ln \left(\frac{W}{560} \right) + 3.18V_w$$

where H is measured in feet, W is measured in grams, and V_w is measured in miles per hour.

4.5 Model Validation

To evaluate the accuracy of the fitted apogee model, the predicted values were compared with the measured apogee values from the successful flight dataset. The prediction error was calculated as

$$\text{Error} = \frac{|H_{pred} - H_{actual}|}{H_{actual}} \times 100\%.$$

Trial	Actual Apogee (ft)	Predicted Apogee (ft)	Error (%)
1	753	759.9	0.91
2	773	763.9	1.18
3	764	756.5	0.98
4	716	727.0	1.53
5	837	847.3	1.23
6	813	800.8	1.50
7	731	732.0	0.14

Table 2: Comparison between measured and predicted apogee values.

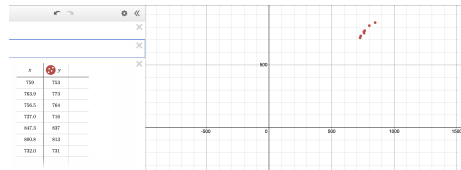


Figure 1: predicted vs actual

The comparison between experimental and predicted apogee values is shown below.

The fitted model demonstrates close agreement with the observed flight data. The prediction error remained below approximately 2% for all tested conditions, indicating that the empirical model captures the dominant relationship between mass, temperature, wind, and apogee within the observed range.

However, the limited number of flight samples restricts the generalization of the model. Additional launches under a wider range of environmental conditions would improve coefficient reliability and reduce uncertainty.

5 Sensitivity Analysis

5.1 First-Order Taylor Approximation

The effect of small variations in launch conditions can be analyzed using a first-order Taylor expansion. Let the apogee model be represented as

$$H = H(W, T, V).$$

Around a reference condition x_0 ,

$$H(x) \approx H(x_0) + \nabla H(x_0)(x - x_0).$$

This approximation describes how small changes in mass, temperature, and wind influence the predicted apogee.

5.2 Gradient-Based Sensitivity

The gradient of the apogee function describes the sensitivity of altitude to each parameter,

$$\nabla H = \begin{bmatrix} \frac{\partial H}{\partial W} \\ \frac{\partial H}{\partial T} \\ \frac{\partial H}{\partial V} \end{bmatrix}.$$

For the proposed model,

$$\frac{\partial H}{\partial W} = \frac{C}{W},$$

$$\frac{\partial H}{\partial T} = B,$$

$$\frac{\partial H}{\partial V} = D.$$

Therefore,

$$\nabla H = \begin{bmatrix} C/W \\ B \\ D \end{bmatrix}$$

The magnitude of each gradient component represents the sensitivity of apogee to a specific variable. The mass term has the largest magnitude, indicating that rocket weight has the strongest influence on altitude within the tested launch conditions.

5.3 Variable Influence Analysis

The magnitude of each gradient component represents the sensitivity of apogee to changes in the corresponding launch parameter.

$$S_W = \left| \frac{\partial H}{\partial W} \right|$$

$$S_T = \left| \frac{\partial H}{\partial T} \right|$$

$$S_V = \left| \frac{\partial H}{\partial V} \right|$$

The sensitivity values can be compared to determine the dominant factor affecting rocket altitude.

$$S_{max} = \max(S_W, S_T, S_V)$$

The parameter associated with the largest sensitivity value has the greatest influence on apogee.

For the fitted model, the mass sensitivity is expected to dominate because the logarithmic mass term produces a larger variation in apogee compared with the temperature and wind terms within the tested flight range.

6 Geometric Interpretation of the Apogee Model

6.1 Second-Order Taylor Expansion

The apogee model can be interpreted as a scalar field over the space of launch parameters,

$$H = H(W, T, V).$$

Let

$$x = \begin{bmatrix} W \\ T \\ V \end{bmatrix}.$$

A second-order Taylor expansion around a reference condition x_0 gives

$$H(x + \Delta x) \approx H(x) + \nabla H(x)^T \Delta x + \frac{1}{2} \Delta x^T \nabla^2 H(x) \Delta x.$$

The first-order term represents sensitivity, while the second-order term represents curvature.

6.2 Curvature of the Apogee Function

The curvature of the apogee model can be described using the Hessian matrix. The Hessian represents the second-order variation of the function with respect to the launch parameters.

$$\nabla^2 H = \begin{bmatrix} \frac{\partial^2 H}{\partial W^2} & \frac{\partial^2 H}{\partial W \partial T} & \frac{\partial^2 H}{\partial W \partial V} \\ \frac{\partial^2 H}{\partial T \partial W} & \frac{\partial^2 H}{\partial T^2} & \frac{\partial^2 H}{\partial T \partial V} \\ \frac{\partial^2 H}{\partial V \partial W} & \frac{\partial^2 H}{\partial V \partial T} & \frac{\partial^2 H}{\partial V^2} \end{bmatrix}.$$

For the proposed model,

$$H = A + B\Delta T + C \ln\left(\frac{W}{W_0}\right) + DV,$$

the second derivatives become

$$\frac{\partial^2 H}{\partial W^2} = -\frac{C}{W^2},$$

while

$$\frac{\partial^2 H}{\partial T^2} = 0,$$

and

$$\frac{\partial^2 H}{\partial V^2} = 0.$$

Therefore, the Hessian is

$$\nabla^2 H = \begin{bmatrix} -\frac{C}{W^2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The nonzero curvature term occurs only in the mass dimension because the model contains a logarithmic mass dependence. This indicates that the influence of mass is nonlinear; the effect of adding additional mass decreases as the rocket becomes heavier.

6.3 Launch Condition Manifold

The set of possible rocket launch conditions can be interpreted as a mathematical manifold. Each point on this space represents a unique combination of physical parameters,

$$x = (W, T, V).$$

The launch condition space is defined as

$$M = \{(W, T, V) \in \mathbb{R}^3\}.$$

The apogee function maps each launch condition to a corresponding altitude,

$$H : M \rightarrow \mathbb{R}.$$

Therefore, the complete rocket model can be viewed as a scalar field over the launch parameter manifold,

$$H = H(x).$$

A small change in launch conditions can be represented by a tangent vector,

$$dx = \begin{bmatrix} dW \\ dT \\ dV \end{bmatrix}.$$

The change in apogee is given by the differential,

$$dH = \nabla H \cdot dx.$$

This represents the local response of rocket altitude to variations in launch conditions.

6.4 Connection and Geodesics

To analyze changes in the launch condition manifold, a connection is introduced to describe how tangent vectors vary across the parameter space.

The covariant derivative is defined as

$$\nabla_i V^k = \partial_i V^k + \Gamma_{ij}^k V^j,$$

where Γ_{ij}^k are the Christoffel symbols representing the connection coefficients.

The Christoffel symbols are given by

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}).$$

They describe how the geometry of the launch condition space changes locally.

The natural paths on the manifold are described by the geodesic equation,

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0.$$

A geodesic represents the smoothest trajectory through the launch parameter space.

Although the regression model predicts apogee directly, the geometric formulation provides a framework for understanding the structure of the parameter space and the interaction between physical variables.

6.5 Application to Flight Optimization

The geometric representation of launch conditions provides a framework for optimizing rocket performance. The apogee function can be viewed as a scalar field over the parameter manifold,

$$H : M \rightarrow \mathbb{R}.$$

The direction of maximum increase in apogee is determined by the gradient,

$$\nabla H.$$

Therefore, an optimization path can be represented as

$$\frac{dx}{dt} = \nabla H(x),$$

where $x = (W, T, V)$ represents the launch condition vector.

In practical rocketry, physical constraints limit the available parameter space. Therefore, optimization occurs on a restricted manifold,

$$M_c \subset M,$$

where constraints such as structural strength, motor capability, and stability requirements are satisfied.

For the proposed model, the gradient indicates how changes in mass, temperature, and wind affect apogee. This allows launch conditions to be adjusted systematically rather than through trial-and-error methods.

7 Conclusion

7.1 Summary of Results

This research developed a mathematical framework for predicting rocket apogee using measurable launch parameters including rocket mass, temperature, and wind conditions.

A model of the form

$$H = A + B\Delta T + C \ln \left(\frac{W}{W_0} \right) + DV$$

was constructed and fitted using experimental flight data through least squares regression.

The resulting model provided a relationship between environmental conditions and rocket performance. Sensitivity analysis using the gradient,

$$\nabla H,$$

showed the relative influence of each parameter on predicted apogee, with rocket mass producing the strongest effect within the tested conditions.

Furthermore, Taylor expansion and geometric interpretations were used to analyze the local behavior of the apogee function, treating launch conditions as a mathematical parameter space.

7.2 Limitations and Future Work

Although the proposed model demonstrates a relationship between launch conditions and apogee, the limited number of experimental flights restricts the accuracy and generalization of the fitted coefficients.

Future improvements include collecting additional flight data, incorporating aerodynamic variables such as drag coefficient and air density, and developing a higher-dimensional nonlinear model.

Further mathematical extensions could include differential equation simulations of rocket trajectories, uncertainty analysis, and more advanced geometric optimization methods for selecting optimal launch conditions.